noc25_ee134

Mock Exam 4 | Course ID: noc25_ee134

Q1. In an AC charging scenario, where does the primary AC-to-DC power conversion take place?

A) Within the external AC EVSE unit
B) Inside the vehicle's Onboard Charger (OBC)
C) At the Central Management System (CMS)
D) Directly at the vehicle's battery terminals

Source: dHc3-Osl9E0 | Difficulty: Easy

Q2. What is the primary motivation for operating DC-DC converters at high switching frequencies?

A) To increase the blocking voltage of the switches
B) To reduce the size and weight of passive components like transformers and inductors
C) To decrease the converter's output voltage ripple
D) To simplify the control loop design

Source: -xXPns3RW-g | Difficulty: Easy

Q3. What is the primary function of a DC-link capacitor in a three-phase AC-DC converter?

A) To increase the modulation index
B) To smooth the DC voltage and absorb ripple currents
C) To generate a high-frequency carrier signal
D) To measure the power factor

Source: -dgKwI8Qoqs | Difficulty: Easy

Q4. What does the term "Unity Power Factor (UPF)" signify in an AC-DC converter?

A) The output DC voltage is equal to the input AC voltage.
B) The current drawn from the AC source is in phase with the voltage.
C) The converter has an efficiency of 100%.
D) The switching frequency is equal to the line frequency.

Source: D3_2zrv4rkY | Difficulty: Easy

Q5. In the context of the Dual Active Bridge (DAB) converter, what does SPS Modulation stand for?

A) Secondary Power Switching Modulation
B) Symmetrical Pulse Shaping Modulation
C) Single Phase Shift Modulation
D) Static Power Supply Modulation

Source: XsicKDSRQhk | Difficulty: Easy

Q6. What is the primary purpose of the Open Charge Point Protocol (OCPP)?

A) To standardize the physical shape of charging connectors
B) To regulate the maximum power output of a charging station
C) To enable interoperability between EVSEs and Central Management Systems (CMS)
D) To define the chemical composition of EV batteries

Source: UAKL2GPPj0U | Difficulty: Easy

Q7. What is the defining characteristic of Discontinuous Conduction Mode (DCM) in a PFC converter?

A) The output voltage is always discontinuous.
B) The inductor current falls to zero during each switching cycle.
C) The switching frequency is always variable.
D) The duty cycle is kept constant at 50%.
Source: FBut4yhA4q0 | Difficulty: Easy

Q8. In a Phase-Shifted Full Bridge (PSFB) converter, what is the purpose of "dead time"?

A) To increase the converter's output power
B) To allow for the charging and discharging of switch output capacitances for ZVS
C) To measure the transformer's leakage inductance
D) To reset the control loop at the end of each cycle
Source: YINqDUxMemQ | Difficulty: Easy

Q9. What is the primary advantage of a Totem Pole PFC converter over a conventional boost PFC converter?

A) It can only operate in Discontinuous Conduction Mode (DCM).
B) It has a higher component count.
C) It eliminates the input diode bridge, reducing conduction losses.
D) It requires a much larger inductor.
Source: X4acp5BiIFs | Difficulty: Easy

Q10. According to the lecture, what is the dedicated digital communication method used in the BEVC-DC001 standard?

A) Power Line Communication (PLC)
B) Wi-Fi
C) Controller Area Network (CAN) Communication
D) Bluetooth
Source: kd8MFLt2h9k | Difficulty: Easy

Q11. In a single-phase full-bridge rectifier with a resistive load, what is the average DC output voltage ($V_{d,avg}$)?

A) $V_{s,peak} / \pi$
B) $V_{s,peak} / 2$
C) $2 * V_{s,peak} / \pi$
D) $V_{s,peak}$
Source: LInjt3L5-IM | Difficulty: Medium

Q12. What is the primary advantage of using a JSON-based implementation of OCPP (OCPP-J) over a SOAP-based one (OCPP-S)?

A) It is more secure by default.
B) It uses XML, which is more structured.
C) It is lightweight, faster, and more scalable due to WebSocket connections.
D) It is only compatible with OCPP version 1.6.
Source: WUYG4Eko3rU | Difficulty: Medium

Q13. In AC Type-2 charging, how does the EV signal to the EVSE that it is ready for charging, causing a transition to State C2?

A) By sending a high-frequency signal over the PP line.
B) By increasing the PWM frequency to 2 kHz.
C) By closing its internal S2 switch, which changes the CP line voltage from +9V to +6V.
D) By drawing a small amount of current from the L1 pin.
Source: 6-xQQwjzPIc | Difficulty: Medium

Q14. What is the primary purpose of introducing an air gap into a filter inductor's core?

A) To increase the inductance value for the same number of turns.
B) To reduce the physical size of the core.
C) To increase the saturation current and make the inductance less dependent on core permeability.
D) To completely eliminate copper losses.
Source: 6X9uRXj0MZQ | Difficulty: Medium

Q15. Why is the output ripple frequency in a full-bridge DC-DC converter (2fsw) twice the switching frequency?

A) Because there are two half-bridges in the converter.
B) Because the output inductor current waveform repeats every half switching cycle (Ts/2).
C) Because the transformer turns ratio is 2:1.
D) Because the output voltage is twice the input voltage.
Source: 38D14JN-3wg | Difficulty: Medium

Q16. What is the main advantage of the DQ rotating reference frame for controlling three-phase AC-DC converters?

A) It increases the switching frequency of the converter.
B) It transforms AC quantities into DC quantities under steady-state, simplifying controller design.
C) It eliminates the need for an output capacitor.
D) It is only applicable to single-phase systems.
Source: 3TF3ZUAokeo | Difficulty: Medium

Q17. In a single-phase boost PFC converter, what is the significance of the "double line frequency ripple"?

A) It improves the power factor of the converter.
B) It is a high-frequency ripple that is easy to filter.
C) It is a low-frequency ripple (e.g., 100/120 Hz) that necessitates a large output capacitor.
D) It only occurs during discontinuous conduction mode.
Source: LGd3GukayGs | Difficulty: Medium

Q18. Why is ZVS turn-on for the lagging leg switches in a Type-2 PWM full-bridge converter difficult to achieve at light loads?

A) The dead time is too long for the lagging leg.
B) The leading leg switches consume all the available energy.
C) The output inductor is short-circuited, so only the small leakage inductance energy is available for capacitor discharge.
D) The lagging leg switches do not have parasitic output capacitances.
Source: XQsOeitdyiE | Difficulty: Medium

Q19. What is the primary purpose of the Signal Level Attenuation Characterization (SLAC) procedure in CCS2 DC charging?

A) To measure the battery's state of charge.
B) To set the charging voltage and current.
C) To logically connect the correct EV and EVSE PLC modems by measuring signal attenuation.
D) To perform an insulation check on the charging cable.
Source: eVsfrIYzqdg | Difficulty: Medium

Q20. In a DC charging scenario, which component acts as the 'master' by dictating the charging profile to the charger?

A) The Supply Equipment Communication Controller (SECC) in the EVSE
B) The vehicle's Onboard Charger (OBC)
C) The vehicle's Battery Management System (BMS)
D) The Central Management System (CMS) of the charging station
Source: a-fvKWBRT_g | Difficulty: Medium

Q21. Why is it advantageous for a balanced three-phase AC-DC converter that its instantaneous input power is constant?

A) It allows the converter to operate without an inductor.
B) It increases the modulation index to above 1.
C) It eliminates the second line frequency harmonic component, allowing for a much smaller output capacitor.
D) It forces the converter to operate in discontinuous conduction mode.

Source: inlEtBllhIE | Difficulty: Medium

Q22. What is the primary role of the Proximity Pilot (PP) pin in an AC Type-2 charging system?

A) To carry the main charging current.
B) To facilitate PWM-based communication for charging states.
C) To provide a ground reference for the vehicle.
D) To detect the plug-in and identify the cable's maximum current capacity via resistance R6.

Source: l4BbXvlP_Rw | Difficulty: Medium

Q23. In the closed-loop control of a single-phase boost PFC converter, why must the inner current loop have a high gain at twice the line frequency (2fs)?

A) To reject the high-frequency switching ripple.
B) To ensure the actual inductor current accurately tracks the 2fs varying reference for unity power factor.
C) To slow down the response of the outer voltage loop.
D) To reduce the required size of the output capacitor.

Source: kGh_kDgqomA | Difficulty: Hard

Q24. According to the lecture, what is the final derived formula for the RMS current flowing through the DC-link capacitor (I_C_RMS) in a three-phase AC-DC converter?

A) I_S_peak * $\sqrt{((3 / \pi) * M * (5/4))}$
B) (3/4) * I_S_peak * M
C) I_S_peak * $\sqrt{((3 / \pi) * M * (5/4) - (9/16) * M^2)}$
D) I_S_peak * $\sqrt{((9/16) * M^2)}$

Source: -dgKwI8Qoqs | Difficulty: Hard

Q25. In the DC CCS2 charging sequence, what specific condition must be met for the EV's internal disconnecting device (contactors) to close during the pre-charging stage?

A) The charging current must exceed 2A.
B) The isolation resistance must be exactly 100 kΩ.
C) The deviation between the EVSE's output voltage and the EV's battery voltage must be less than 20V.
D) The EVSE must confirm the connector is locked.

Source: cEaVn-3aSNk | Difficulty: Hard

Q26. During the design of a filter inductor, what is the fundamental inequality derived from the four governing equations that is used for core selection?

A) $AC * WA / MLT \geq (\rho * L * I_{max}) / (B_{max} * R * ku)$
B) $AC^2 * WA / MLT \geq (\rho * L^2 * I_{max}^2) / (B_{max}^2 * R * ku)$
C) $AC * WA^2 / MLT \geq (\rho * L * I_{max}) / (B_{max} * R * ku)$
D) $AC^2 * WA / LC \geq (\rho * L^2 * I_{max}^2) / (B_{max}^2 * R * ku)$

Source: 6X9uRXj0MZQ | Difficulty: Hard

Q27. In a Totem Pole PFC converter, what causes the negative inductor current spike that contributes to zero-crossing distortion?

A) A delay in the turn-on of switch S4, causing the inductor voltage to become approximately V_o.
B) The body diode of S3 conducting for too long.
C) When S1 turns on for a short duration while vs is near zero, applying a large negative voltage (vs - V_o) across the inductor.
D) The output capacitor discharging back into the inductor.

Source: RxJCQteGJQA | Difficulty: Hard

Q28. To guarantee DCM operation in a flyback-based PFC converter under all conditions, the primary inductance L must be selected to be less than the minimum critical inductance (L_critical_min). How is this L_critical_min calculated?

A) At maximum load resistance (RL_max) and maximum input voltage (Vs,peak_max).
B) At minimum load resistance (RL_min) and maximum input voltage (Vs,peak_max).
C) At maximum load resistance (RL_max) and minimum input voltage (Vs,peak_min).
D) At minimum load resistance (RL_min) and minimum input voltage (Vs,peak_min).

Source: YYOndlU0fVE | Difficulty: Hard

Q29. In the closed-loop control of a three-phase AC-DC converter, how is the 'K' factor, which links the converter's output current to the d-axis current, derived?

A) By using pole-zero cancellation in the current loop.
B) From the bandwidth of the voltage sensor.
C) From the principle of power balance between the three-phase input and the two-phase DQ frame representation.
D) By measuring the parasitic inductance of the switches.

Source: KmgcXbLuwno | Difficulty: Hard

Q30. What is the primary reason a forward converter has limited core utilization compared to a full-bridge converter?

A) It uses a larger air gap in its transformer core.
B) It operates at a much lower switching frequency.
C) It only utilizes one quadrant of the transformer's B-H curve (0 to Bmax).
D) It requires a lossy reset mechanism.

Source: TlKIeTe3Fy0 | Difficulty: Hard

Answer Key

1: B	2: B	3: B	4: B	5: C
6: C	7: B	8: B	9: C	10: C
11: C	12: C	13: C	14: C	15: B
16: B	17: C	18: C	19: C	20: C
21: C	22: D	23: B	24: C	25: C
26: B	27: C	28: D	29: C	30: C